

# Chapter: Getting to Know Plants 1.

## Classification of Plants Based on Growth Habits

If you look closely at the plants around you, they aren't just different in color—they vary drastically in height, thickness of stems, and lifespans. We classify them into five distinct structural categories:

### A. Herbs

- **Characteristics:** Very short plants with soft, green, and tender stems. They usually do not have many branches.
- **Lifespan:** Short (often living for just one season).
- **Examples:** Tomato, Mustard, Paddy, Wheat, Coriander, Mint.

### B. Shrubs

- **Characteristics:** Medium-sized plants that develop branches near the very base of the stem. The stem is hard but not exceptionally thick.
- **Examples:** Lemon, Rose, China Rose (Hibiscus), Jasmine.

### C. Trees

- **Characteristics:** Very tall plants with a thick, hard, woody brown stem called the **trunk**. The branches develop high up on the trunk, far above the ground.
- **Examples:** Mango, Neem, Banyan, Coconut.

### D. Creepers & Climbers

- **Creepers:** Plants with weak stems that cannot stand upright and instead spread horizontally along the muddy ground (e.g., Pumpkin, Watermelon).
- **Climbers:** Weak-stemmed plants that use neighboring structures (like walls, sticks, or other trees) to support themselves and climb upward (e.g., Money Plant, Pea Plant, Grapevine).

## 2. The Leaf: The Food Factory of the Plant

The leaf is a flat, lateral structure attached directly to the stem. It functions as the respiratory and digestive hub of the plant.

### Structural Parts of a Leaf:

- **Lamina (Leaf Blade):** The broad, green, flat surface area of the leaf.
- **Petiole:** The tiny, stalk-like structure that joins the lamina to the main stem.
- **Midrib:** The central thick vein running right down the middle of the leaf from base to tip.
- **Veins:** Fine, thread-like pipelines branching out from the midrib to transport water and nutrients.

### Leaf Venation (Arrangement of Veins)

The design made by veins inside a leaf is called **leaf venation**. It comes in two distinct types:

1. **Reticulate Venation:** Veins form a net-like web pattern on both sides of the midrib. Found in **dicot plants** (plants with seeds containing two halves). *Examples: Rose, Coriander, Mango, Peepal.*
2. **Parallel Venation:** Veins run completely straight and parallel to one another from the base to the leaf tip. Found in **monocot plants** (seeds with one whole piece). *Examples: Grass, Banana, Maize, Wheat.*

## Two Core Functions of Leaves:

- **Transpiration:** The process by which plants release excess internal water into the air in the form of water vapor through microscopic pores called **stomata**. This creates a cooling effect and draws more water up from the roots.
- **Photosynthesis:** The chemical process by which green leaves prepare food for the entire plant. Leaves utilize carbon dioxide from the air and water from the soil in the presence of green **chlorophyll** pigment and sunlight.

## 3. The Root System: The Foundation

Roots are the underground, non-green parts of a plant that grow downward into the dark soil.

| Property            | Taproot System                                                                                           | Fibrous Root System                                                                                             |
|---------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Structure           | Features one deep, thick main root (primary root) with smaller side roots (lateral roots) branching out. | Has no main single root. A cluster of thin, identical-sized roots sprouts altogether from the base of the stem. |
| Associated Venation | Plants with taproots always have <b>reticulate venation</b> leaves.                                      | Plants with fibrous roots always have <b>parallel venation</b> leaves.                                          |

|                 |                                          |                                               |
|-----------------|------------------------------------------|-----------------------------------------------|
| <b>Examples</b> | Carrot,<br>Radish,<br>Mustard,<br>Mango. | Grass,<br>Wheat,<br>Maize,<br>Rice,<br>Onion. |
|-----------------|------------------------------------------|-----------------------------------------------|

### Main Functions of Roots:

1. **Anchorage:** They anchor the plant firmly into the loose soil, preventing it from being blown away by high winds.
2. **Absorption:** They suck up vital groundwater and dissolved minerals.
3. **Soil Conservation:** They hold soil particles tightly together, preventing topsoil erosion.

## 4. The Stem: The Highway System

The stem forms the main axis of the shoot system, growing upward toward the sunlight.

- **Conduction Pathway:** The stem acts like a two-way street pipeline. It contains specialized interior channels that conduct water and mineral salts upward from the roots to the leaves, and distributes prepared starches from the leaves down to the roots and storage organs.
- **Nodes & Internodes:** The exact point on a stem where a new leaf sprouts is called a **node**. The empty segment of stem between two consecutive nodes is called an **internode**.

## 5. Structure of a Flower

The flower is the most attractive and colorful reproductive part of a flowering plant.

As illustrated in the cross-section above, a classic flower consists of four main concentric rings (whorls):

### A. Sepals (Outer Ring)

Small, green, leaf-like structures forming the outermost layer. They protect the delicate inner floral parts while the flower is still a closed, vulnerable bud.

### B. Petals (Second Ring)

The broad, bright, colorful, sweet-smelling parts of the flower. Their main evolutionary job is to look and smell attractive to draw insects like bees and butterflies for **pollination**.

### C. Stamens (Male Reproductive Part)

Located inside the petals. Each individual stamen is broken down into two components:

- **Anther:** A small, swollen, double-lobed sac at the top that produces powdery golden **pollen grains**.
- **Filament:** The long, slender supporting stalk holding the anther up.

### D. Pistil / Carpel (Female Reproductive Part)

The innermost, vase-shaped central structure of the flower. It is split into three parts:

- **Stigma:** The sticky top platform surface that catches landing pollen grains.
- **Style:** The long, hollow neck tube that pollen travels down.
- **Ovary:** The swollen, bulging bottom chamber. The ovary contains tiny seed-like structures called **ovules**. After fertilization, the ovary grows to become the **fruit**, while the ovules inside transform into **seeds**.

# Multiple Choice Questions (Practice Test)

**Q1. Which of the following plants has leaves with parallel venation?**

- A) Rose
- B) Peepal
- C) Maize
- D) Tomato

**Q2. The process by which water escapes from the surface of leaves in the form of vapor is called:**

- A) Photosynthesis
- B) Transpiration
- C) Respiration
- D) Conduction

**Q3. You observe a weed plant with a fibrous root system. Without looking at its leaves, you can safely predict that its leaf venation will be:**

- A) Reticulate venation
- B) Parallel venation
- C) Missing entirely
- D) Irregular net-like pattern

**Q4. Which specific part of a flower converts into a protective fruit after fertilization takes place?**

- A) Ovary
- B) Ovule
- C) Anther
- D) Petal

**Q5. Mint and coriander plants are classic examples of:**

- A) Shrubs
- B) Trees
- C) Climbers
- D) Herbs

### **Answer Key & Explanations**

1. **(C) Maize** — Monocot grass family crops like maize, wheat, and rice feature long, slender leaves with veins running parallel to each other.
2. **(B) Transpiration** — Transpiration releases excess water via leaf stomata, which creates structural suction pressure to pull new nutrients up from the soil.
3. **(B) Parallel venation** — Nature follows a strict relationship link: Fibrous roots always coordinate with parallel venation, while taproots coordinate with reticulate venation.
4. **(A) Ovary** — The outer ovary wall swells up to become the flesh of the fruit, while the inner microscopic ovules harden into the seeds.
5. **(D) Herbs** — Both mint and coriander have soft, delicate, green stems with zero woody properties and a very short seasonal lifecycle.